8A
QUANTUM MODEL OF ATOM

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

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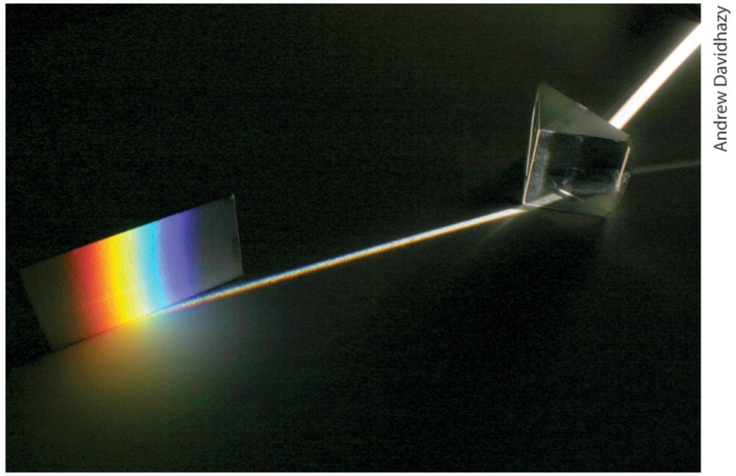
LEARNING OBJECTIVES

CHAPTER 8

1. Describe and apply the dual nature of light.
2. Calculate energy, frequency, or wavelength of light.
3. Use the Bohr model of energy levels in atoms to explain light emission and absorption by gaseous atoms.
4. Calculate the energy or wavelength of a specific electronic transition in a hydrogen atom.
5. Visualize and describe the spatial distributions and energies of atomic electrons using the quantum model of the atom.
6. Use the concept of quantum numbers to characterize the energies, spatial distributions, and spins of atomic electrons.
7. Sketch the electronic energy levels of an atom.
8. Add electrons to atomic orbital energy (energy level) diagrams.
9. Determine electron configurations for the elements using the concepts of shells, subshells, and orbitals.

A Brief Exploration of Light

8.1



Historically, light has been treated as a ***wave***.

- Two components: electric field + magnetic field
- Explains wave-based observable phenomena:
  - *Reflection*: Alter direction off surface
  - *Diffraction*: Alter direction through opening
  - *Refraction*: Alter direction through medium

Electromagnetic (EM) Spectrum: The full range of "light" energies (also called radiation).

Visible light: A more accurate term for what we usually refer to simply as "light."

A Brief Exploration of Light

8.1

Central characteristics of a wave:

- Amplitude: Size, height, or intensity of a wave
- Wavelength: Distance between two equivalent points on a wave
- Frequency: Number of waves passing through a point per unit time

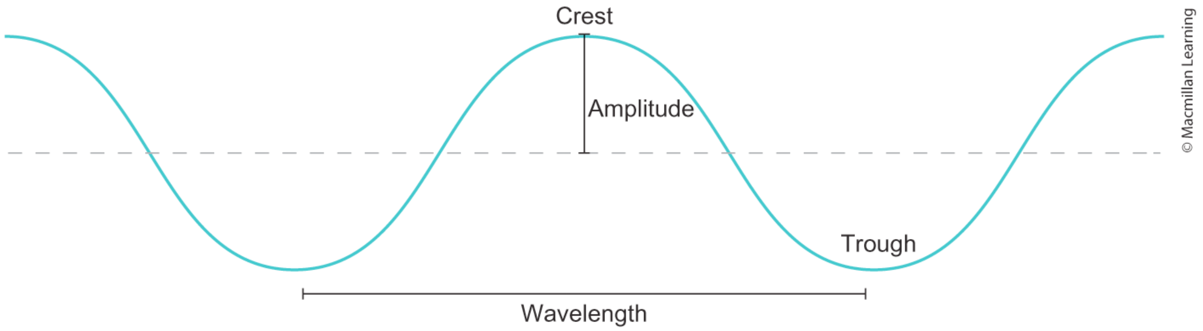


FIGURE 8.2

A Brief Exploration of Light

8.1

Frequency (ν) and wavelength (λ) are related through the **speed of light** (*c*).

- There is an ___________ relationship between ν and λ.
- The energy of a wave is ___________ related to λ and ___________ related to ν.

______________________________ ______________________________

A Brief Exploration of Light

8.1

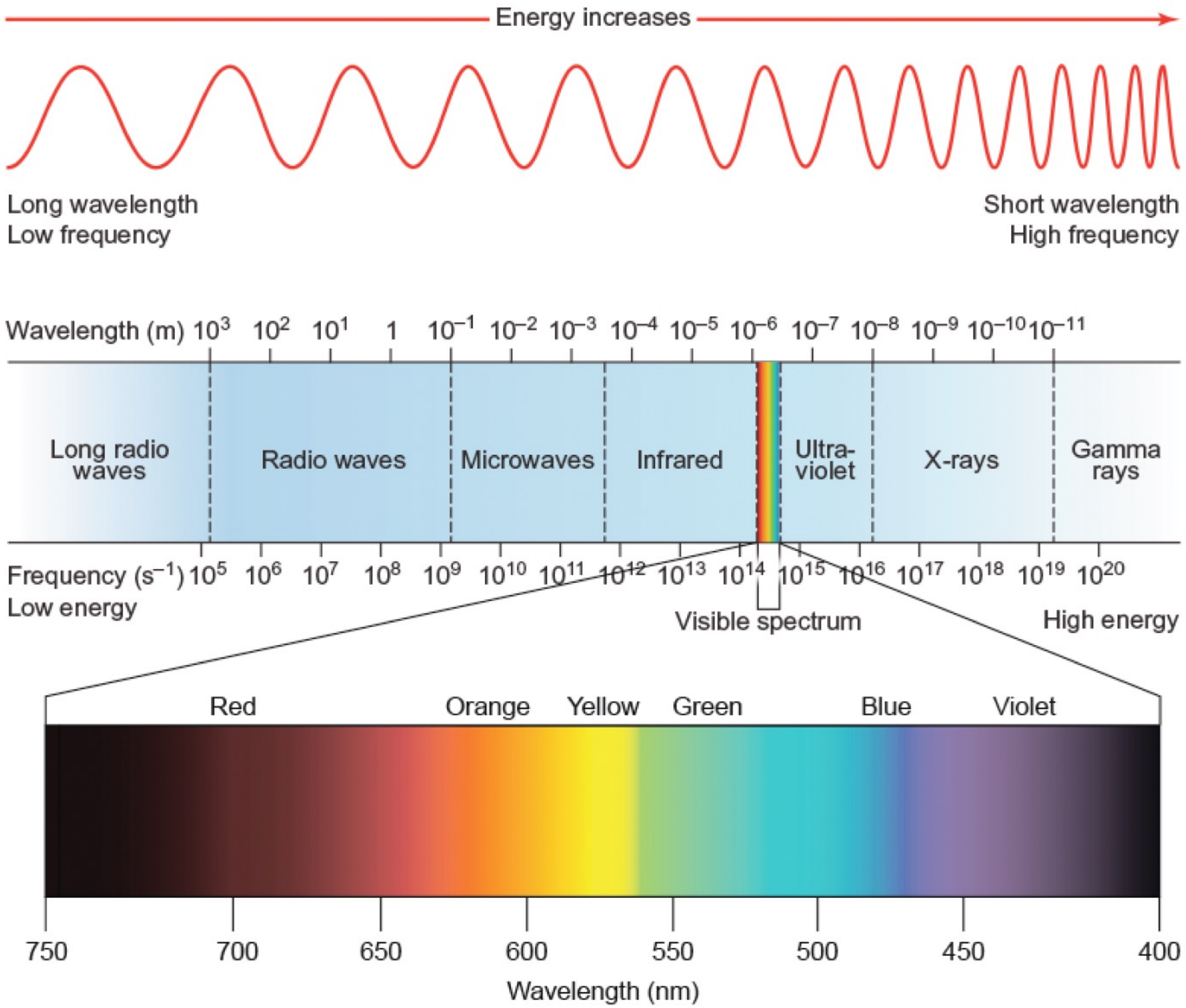


FIGURE 8.4
ELECTROMAGNETIC SPECTRUM

A Brief Exploration of Light

8.1

Until the early 1900s, ***light was treated only as a <u>wave</u>***, which was validated by the observation of diffraction patterns in the "double slit experiment."

However, the results of later experiments (the photoelectric effect) led to the necessity for the treatment of ***light as a <u>particle</u>***.

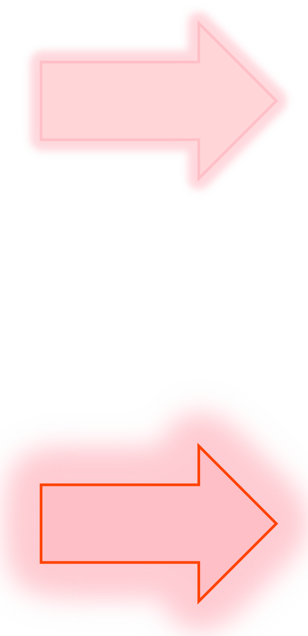
This led to the conception of <u>wave-particle duality</u>:

A Brief Exploration of Light

8.1

Einstein proposed that light be described as a collection of **photons** or *wave-particles*.

- Each photon contains a ***packet of energy***.
- Bright light has *many* photons, and dim light has *few* photons.



Energy of a single photon of light:

Energy of N photons of light:

EXAMPLE PROBLEM

Learning Objective(s): 8.2

What is the energy (in J) of a *single* photon of blue light of frequency 6.4×10^{14} Hz?

What is the energy (in J) of *one mole* of photons of the same frequency?

IN-CLASS QUESTION 1

Learning Objective(s): 8.1, 8.2

A laser pulse of light with wavelength 532 nm contains 0.00467 J of energy.

How many total photons are in this laser pulse?

The Bohr Model of the Atom

8.2

Light absorption: An atom in its **ground state** (*the lowest energy state*) absorbs light. The atom gains energy and exists in an **excited state** (*a higher energy state*).

Light emission: An atom in an *excited (higher energy)* state *releases* or *gives off* its *excess* energy by emitting photons of light.

The Bohr Model of the Atom

8.2

Each atom will *absorb* and *emit* characteristic (specific) wavelengths or colors of light that are unique to that element.

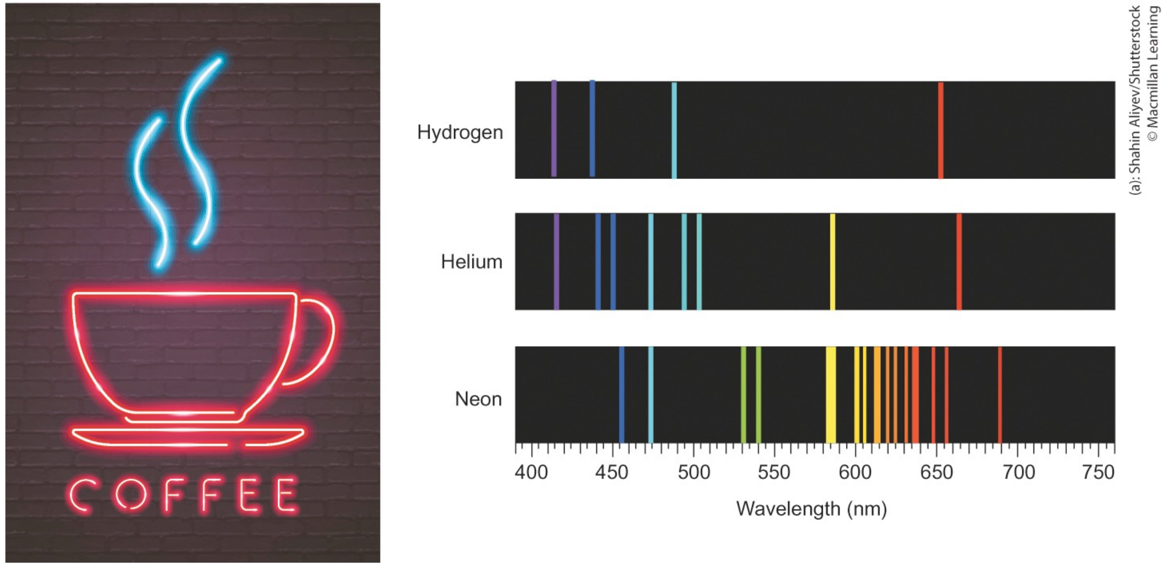


FIGURE 8.11

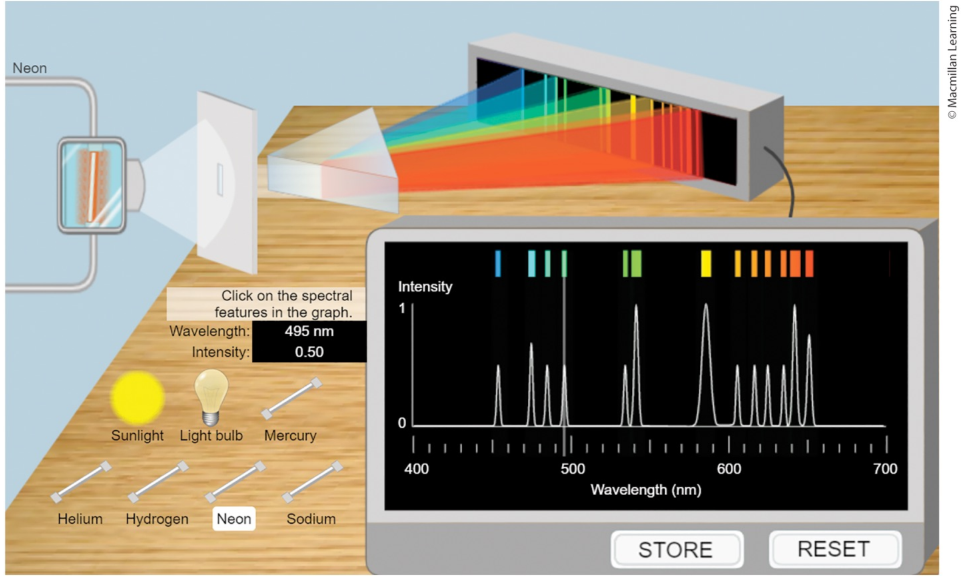


FIGURE 8.12 Spectroscopy App

Atomic spectroscopy can be used to identify elements because they have characteristic spectra due to their unique electronic energy levels.

The Bohr Model of the Atom

8.2

Bohr Model of the Atom:

- Proposed by Niels Bohr (Nobel Prize 1922)
- When gaseous atoms of a given element are heated, they emit *only* specific wavelengths of light—an emission spectrum.

The Bohr Model of the Atom

8.2

The energy level *closest* to the nucleus is *lowest in energy* and called the ground state.

- An absorption spectrum is observed when electrons absorb energy and move to higher energy levels called excited states.

- An emission spectrum is observed when electrons in excited states emit energy to return to lower energy states.

The energy of the light absorbed or emitted directly correlates to the difference between the existing energy levels (called orbits) in the atom.

The Bohr Model of the Atom

8.2

Rydberg equation: The mathematical equation that predicts the wavelengths of all hydrogen absorption/emission lines.

Energy of orbit/state n:

Energy of photon absorbed/emitted upon transition between two orbits/states, n_1 and n_2:

The Bohr Model of the Atom

8.2

Electronic energy levels and possible electronic transitions in the hydrogen atom.

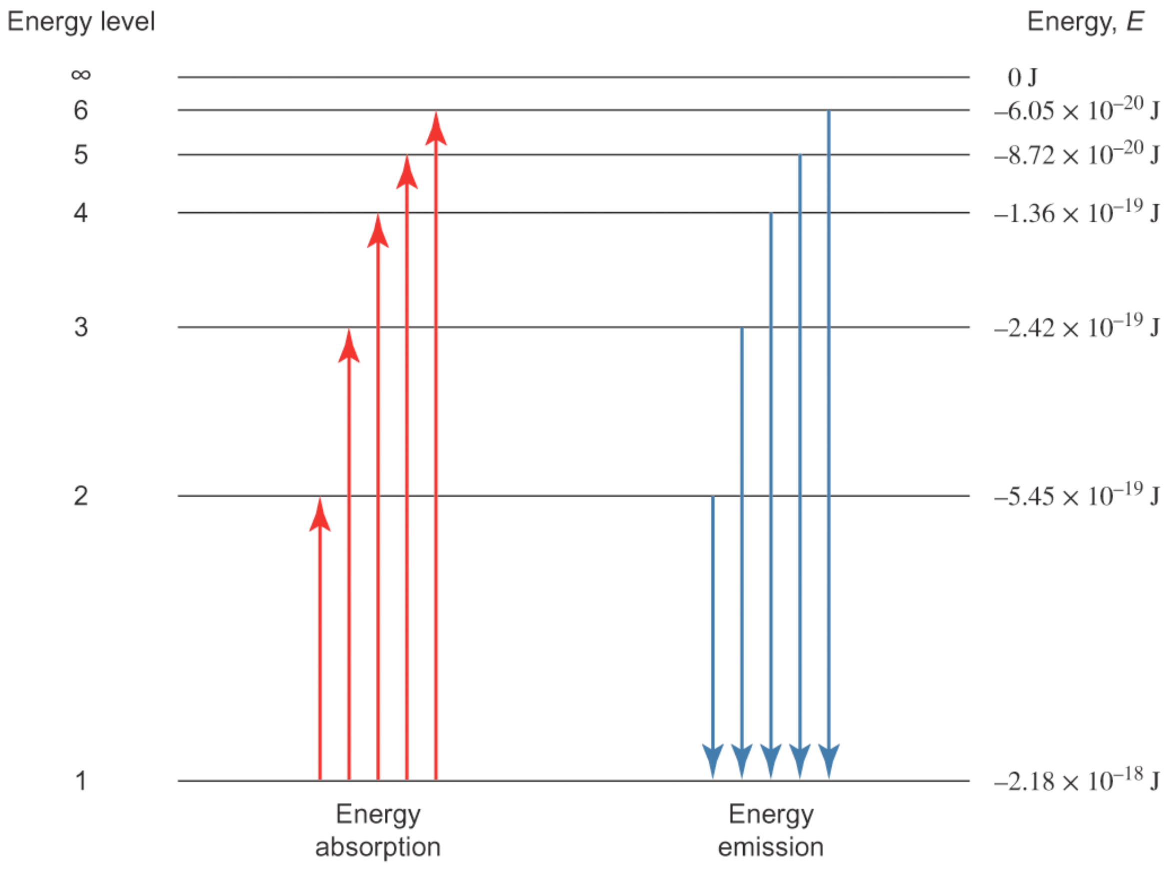


FIGURE 8.13

(See notes on next slide.)

The Bohr Model of the Atom

8.2

Notes on electronic energy levels (E_n)

- The biggest ΔE for any two *adjacent* energy levels (n) in any atom is always between the first and second energy levels (E_1 and E_2).

- The spacing between adjacent energy levels (ΔE) gets smaller as n increases (or as the distance of the orbit from the nucleus increases).

- A completely detached or removed electron is defined as $n = \infty$ (infinity) and assigned a value of $E_\infty = 0$ J.

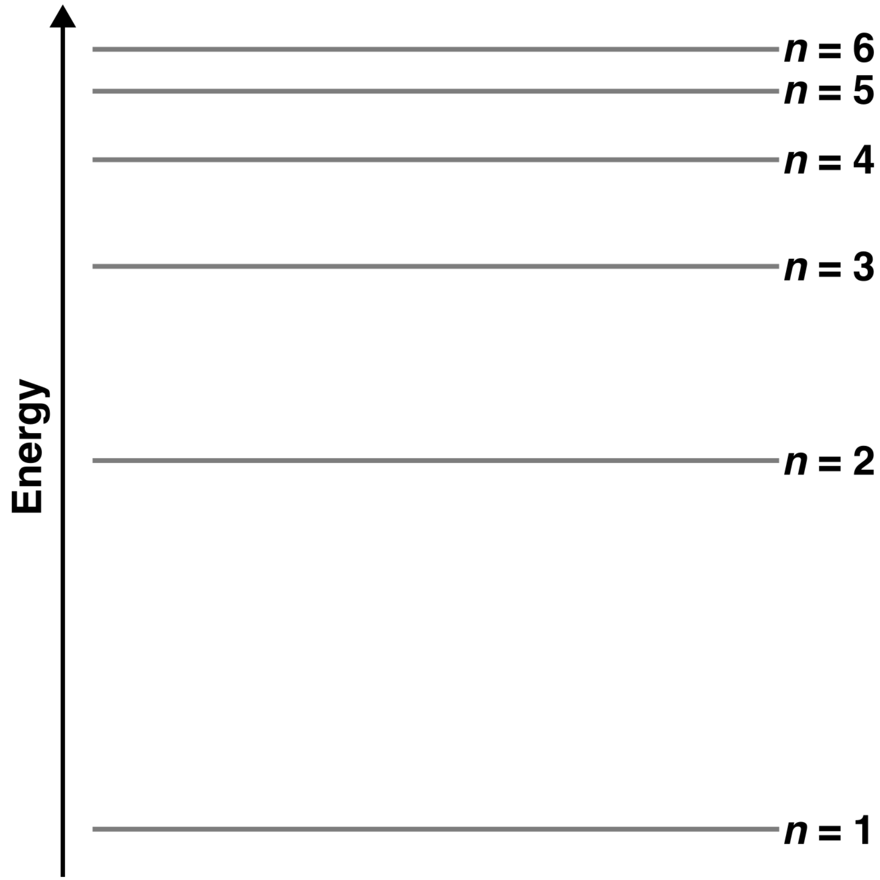
- If $E_{n=\infty} = 0$ J, *all* other E_n are given negative values as they are more stable (i.e., lower in energy) than the detached electron in the highest energy level ($n = \infty$).

IN-CLASS QUESTION 2

Learning Objective(s): 8.3

For an electron in the hydrogen atom, identify the transitions that will result in the emission of a photon of (i) the ***shortest wavelength*** and (ii) the ***lowest frequency***?

- ❑ n = 6 to n = 5
- ❑ n = 5 to n = 4
- ❑ n = 4 to n = 3
- ❑ n = 3 to n = 2
- ❑ n = 2 to n = 1



EXAMPLE PROBLEM

Learning Objective(s): 8.4

Calculate the wavelength (in nm) of light *absorbed* when the electron in a hydrogen atom is excited from the n = 2 shell to the n = 5 shell.

BONUS QUESTION

Assessed on: *Undetermined*

Which of the following statements are true regarding the energy diagram below?

Note: Lines 1–5 (labeled on the diagram) represent electron transitions.

- ❑ Lines 1 and 3 represent emissions and Lines 2, 4, and 5 represent absorptions.
- ❑ The transition associated with the longest wavelength is Line 2.
- ❑ The frequency associated with Line 5 is greater than that associated with Line 4.
- ❑ The wavelength associated with Line 3 is shorter than that associated with Line 1.
- ❑ The photon of light associated with Line 4 has more energy than that associated with Line 1.